

IDLE — no transaction

0x00

a READ arrives:
write the identity

LIVE — nothing pending yet

0xC0 . . 0xCF

top bit SET; low nibble = the DNP3 sequence
a token sees difference 0x00 → still mine

the first response is admitted:
add 0x50 (one-shot)

LIVE — a response is queued behind the ACK

0x10 . . 0x1F

top bit CLEAR, and never 0x00
a token sees difference 0xB0 → STILL mine

ACK released,
nothing pending

the queued response
leaves

Three disjoint domains in ONE byte. The live states differ only in the top bit, so the switch separates them with a single signed comparison against zero.